ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

BAG OF WORDS CLASSIFIER

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

A TYPICAL TEXT MINING PIPELINE (Recap)

Try to make texts "better formatted"

Make the text data machine-actionable

Specifically for the end-task

Bag-of-words Text Classifier

- What's text classification? How to evaluate a classifier?
- Bag-of-words Representations
- Building a linear classifier upon bag-of-words

Bag-of-words Text Classifier

- **What's text classification? How to evaluate a classifier?**
- Bag-of-words Representations
- Building a linear classifier upon bag-of-words

Given a review document, can we estimate its sentiment from its content?

Sherlockian

Love the Switch - A Casual Gamer's Best Console

Reviewed in the United States on September 6, 2019

Style: Console | Color: Gray HAC-001(-01) | Verified Purchase

This is my review of the 'V2' (2019) version of the Nintendo Switch (model #HAC-001(-01)).

So a little background, I was a enthusiastic PC gamer in college who built my own PCs for the purpose and am now - due to the demands of adulthood - someone who only plays the occasional game. My last handheld Nintendo console was the black and white Game Boy Pocket in the early 90's, so I've been away from the platform ---- a while.

I bought the Switch purely because I learned they had ported Skyrim to it, and as a huge fan of the Elder Scrolls genre I was hoping to devote some of my train commute to questing around Tamriel. In this sense, the Switch does not disappoint. I am not a graphics connoisseur, but - compared to the PC version - Skryim plays really well on the Switch. No perceptible lag, colors are crisp. Yes, it's missing the expansions and yes, the text can be a little small, but that's a fair trade-off for the portability this platform brings.

So far, I only own 2 games - Skyrim and MarioKart 8 (another throwback to my college dorm days on the N64), and I love playing both on the Switch. MaroKart is obviously the newer game and so the graphics and gameplay are

Read more

456 people found this helpful

DWBelfils

The device is great. The everything else is not.

Reviewed in the United States on August 23, 2019

Style: Console | Color: Gray HAC-001(-01) | Verified Purchase

Everything about the physical device, i feel like is pretty well made. Good set up, some minor future issues that are mitigated by screen protectors and similar. Several easy to work with ways to play it, kinda fun in its novelties on its build.

My issue?
1) games are ridiculously expensive. So many are over priced.
2) Online play required by Nintendo for a not good service is poor. Especially for things like minecraft.
3) Where are any older games. The games that Nintendo has no need to keep out of the store. The old pokemon, Zelda games, or any games really. Like, great you are putting witcher 3 on there...But not Orignal Zelda and Ocherina of time?

Physical? Great! Nintendo online and games? Bad. 1 or 2 games does not make the switch worth buying. PC has better prices, easier to work with, and you can play almost any game out there.

941 people found this helpful

SENTIMENT ANALYSIS AS AN EXAMPLE

Sentiment Analysis on the IMDB dataset

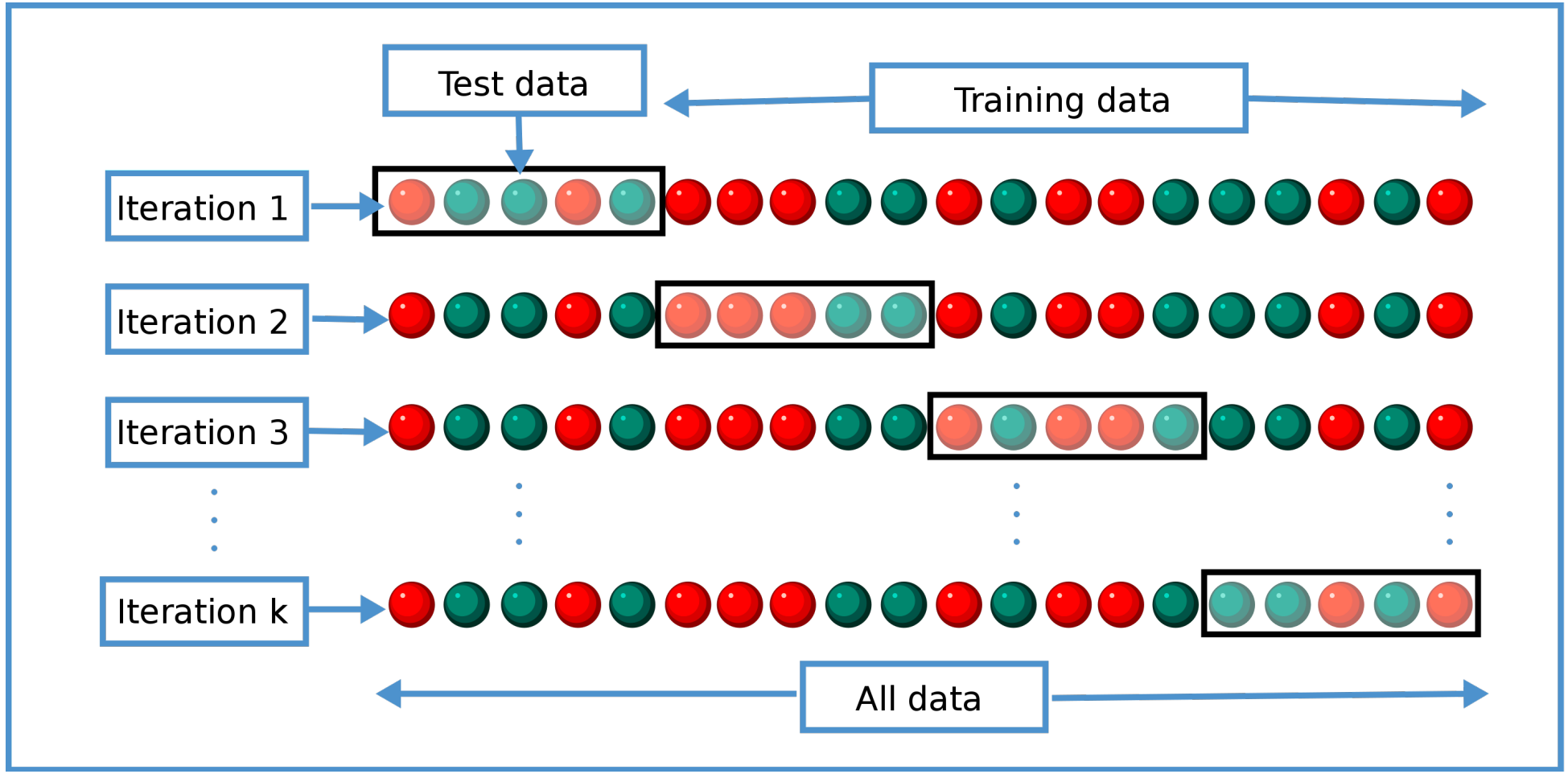
In [1]:
```
import pandas as pd

df = pd.read_csv('movie_data.csv')

print(len(df))

df.head(10)
```

```
50000
```

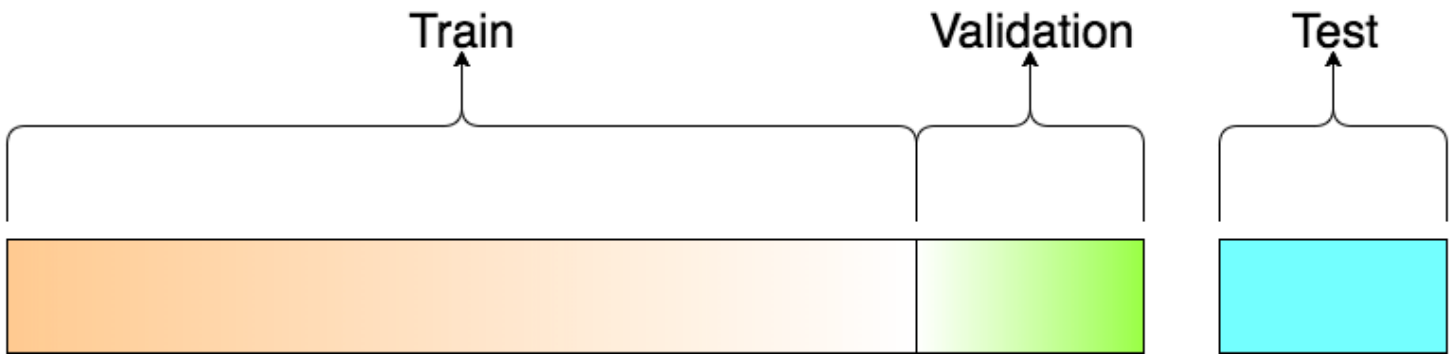
Out[1]:

	review	sentiment
0	In 1974, the teenager Martha Moxley (Maggie Gr...	1
1	OK... so... I really like Kris Kristofferson a...	0
2	***SPOILER*** Do not read this, if you think a...	0
3	hi for all the people who have seen this wonde...	1
4	I recently bought the DVD, forgetting just how...	0
5	Leave it to Braik to put on a good show. Final...	1
6	Nathan Detroit (Frank Sinatra) is the manager ...	1
7	To understand "Crash Course" in the right cont...	1
8	I've been impressed with Chavez's stance again...	1
9	This movie is directed by Renny Harlin the fin...	1

K-fold Cross Validation



Train/Dev/Test



Evaluation Metrics

Predicted Values	Actual Values: Positive (1)	Actual Values: Negative (0)
Positive (1)	TP	FP
Negative (0)	FN	TN

- TP: True Positive
  - Predicted as Positive & True Label Positive
- TN: True Negative
  - Predicted as Negative & True Label Negative

- Similarly, we have …

- FP: False Positive
- FN: False Negative

Evaluation Metrics: Accuracy, Precision, Recall, and F1

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

- **Acc**uracy
  - (TP + TN) / (TP + FP + FN + TN)
- **Prec**ision
  - TP / (TP + FP)
- **Rec**all
  - TP / (TP + FN)

- F1
  - Harmonic Mean between Prec and Rec
  - 2 * Prec * Rec / (Prec + Rec)